Performance Testing

Date	13 November 2026
Team ID	0157CS241131
Project Name	AI Travel Itinerary Planner (Trip Genius AI)
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	e.g. Locust / k6 / Postman
Type of Testing	Load Testing (planned)
Target Module / API	GET /health; GET /api/trips; POST /api/itineraries/generate
Test Environment	Hosted (Render backend + Vercel frontend)
Test Date	TO ADD

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Health check endpoint under light load	10	60	All requests return 200, avg < 2 s
2	List saved trips (authenticated)	10	60	200 OK, only own trips returned
3	Generate 3-day itinerary (Groq call)	3	120	Valid JSON itinerary; errors < 1% (rate limits may apply)
4	Mixed browsing and generation	20	180	No crashes; graceful errors if AI rate-limited

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	TO ADD	TO ADD	Non-AI endpoints; AI calls tracked separately
2	Response Time (Max)	< 5 seconds	TO ADD	TO ADD	
3	Throughput (Req/sec)	-	TO ADD	TO ADD	
4	Error Rate	< 1%	TO ADD	TO ADD	
5	CPU Utilization	< 80%	TO ADD	TO ADD	Render dashboard

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
6	Memory Utilization	< 80%	TO ADD	TO ADD	Render dashboard

Step 4: Observations & Analysis

TO ADD after running the tests Note: generation latency depends mainly on the Groq model response time; the free Render tier may add a cold-start delay on the first request.

Step 5: Screenshots / Evidence

Insert tool screenshots here